FLATTENING A STACK BY AN ADMISSIBLE BLOWUP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2115-2129 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 2131-2178, the complete author proof (46 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 33 of queue_stacks-stacks-more-morphisms.json (pb=1634, stmt 2115-2129, proof 2131-2178).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04T5: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2178–2178. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0865: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2178–2178. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 087E: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2178–2178. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0507: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2178–2178. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : \mathcal{X} \to Y$ be a morphism from an algebraic stack to an algebraic space. Let $V \subset Y$ be an open subspace. Assume*

(1) *Y is quasi-compact and quasi-separated,*
(2) *f is of finite type and quasi-separated,*
(3) *V is quasi-compact, and*
(4) *$\mathcal{X}_V$ is flat and locally of finite presentation over V.*

Then there exists a V-admissible blowup $Y' \to Y$ and a closed substack $\mathcal{X}' \subset \mathcal{X}_{Y'}$ with $\mathcal{X}'_V = \mathcal{X}_V$ such that $\mathcal{X}' \to Y'$ is flat and of finite presentation.

Proof. Observe that $\mathcal{X}$ is quasi-compact. Choose an affine scheme U and a surjective smooth morphism $U \to \mathcal{X}$. Let $R = U \times_{\mathcal{X}} U$ so that we obtain a groupoid (U, R, s, t, c) in algebraic spaces over Y with $\mathcal{X} = [U/R]$ (Algebraic Stacks, Lemma 04T5). We may apply More on Morphisms of Spaces, Lemma 087E to $U \to Y$ and the open $V \subset Y$. Thus we obtain a V-admissible blowup $Y' \to Y$ such that the strict transform $U' \subset U_{Y'}$ is flat and of finite presentation over Y'. Let $R' \subset R_{Y'}$ be the strict transform of R. Since s and t are smooth (and in particular flat) it follows from Divisors on Spaces, Lemma 0865 that we have cartesian diagrams

$$\begin{array}{ccc} R' & \longrightarrow & R_{Y'} \\ \downarrow & & \downarrow{\scriptstyle s_{Y'}} \\ U' & \longrightarrow & U_{Y'} \end{array} \quad \text{and} \quad \begin{array}{ccc} R' & \longrightarrow & R_{Y'} \\ \downarrow & & \downarrow{\scriptstyle t_{Y'}} \\ U' & \longrightarrow & U_{Y'} \end{array}$$

In other words, U' is an $R_{Y'}$-invariant closed subspace of $U_{Y'}$. Thus U' defines a closed substack $\mathcal{X}' \subset \mathcal{X}_{Y'}$ by Properties of Stacks, Lemma 0507. The morphism $\mathcal{X}' \to Y'$ is flat and locally of finite presentation because this is true for $U' \to Y'$. On the other hand, we already know $\mathcal{X}' \to Y'$ is quasi-compact and quasi-separated (by our assumptions on f and because this is true for closed immersions). This finishes the proof. □